MATH 128a: Homework #1

Instructor: F. Weber

Due by: **Tuesday, September 9, 2025, 11:59pm**

Problem 1: Preliminaries

- Read the syllabus for the course and answer the following questions:
  (a) Will there be a final exam for this class? If so, when is it?
  (b) How many programming assignment scores will be dropped for the computation of the grade?
  (c) Will there be quizzes in this class? If so, how many/how often?
  (d) Where can you find announcements and materials related to the course?
  (e) Where should you ask math related questions for this course?

Answer of Problem 1

(a) Yes, December 16 from 3-6pm

(b) One

(c) Yes, six quizzes about biweekly

(d) bcourses

(e) in office hours or on Ed Discussion

Problem 2: Problems from textbook Section 1.1

Solve problems 1b, 5b(4b), 8(6), 11abcd(9abcd), 21, 24ac(26ac), 28abc(27abc), 29bc(28bc) from Section 1.1 of the textbook.
If the problem has a different number in edition 9 of the textbook, it is in parentheses.
For problem 11b: express the error bound as a function of x.
For problem 21: find a bound independent of x that works for all x in the given range.

Answer of Problem 2

- **1b:** Let $f(x) = (x-2)^2 - \ln x$. We have $f(1) = 1$ and $f(2) = -\ln 2$. The zero is between 1 and $-\ln 2$, so by the intermediate value theorem there is a real number $c \in [1, 2]$ with $f(c) = 0$.

  For the interval $[e, 4]$, we have $f(e) = (e-2)^2 - 1$ and $f(4) = 2^2 - \ln 4$. Since $e - 2 < 1$, we have $f(e) < 0$ and since $\ln 4 < 4$ we have $f(4) > 0$. Again the desired result follows from the intermediate value theorem.

- **5b**: The denominator is non-zero on $[0.5, 1]$, so $f \in C^1[0.5, 1]$. As noted in class, we just have to check extrema and zeros of f to find extrema of $|f(x)|$. The derivative of f is

$$\begin{aligned} f'(x) &= \frac{4(x^2 - 2x) - (4x - 3)(2x - 2)}{(x^2 - 2x)^2} \\ &= \frac{4x^2 - 8x - 8x^2 + 8x + 6x - 6}{(x^2 - 2x)^2} \\ &= \frac{-4x^2 + 6x - 6}{(x^2 - 2x)^2}. \end{aligned}$$

  The discriminant $6^2 - 4(4)(6)$ is negative, so the numerator has constant sign (which is seen to be negative by checking $x = 1$ for instance). Thus, $f'(x)$ is negative on $[0.5, 1]$ and the extrema of $f(x)$ occur at the endpoints: $f(0.5) = -1/(-.75) = 4/3$, and $f(1) = 1/(-1) = -1$. The only zero of $f(x)$ is at $x = 3/4$. So the minimum and maximum values of $|f(x)|$ are $|f(3/4)| = 0$ and $|f(0.5)| = 4/3$.

- **8**: We argue by contradiction. Suppose to the contrary that there are two points $p, q \in [a, b]$ with $f(p) = f(q) = 0$. Since f is differentiable on (a, b), Rolle's theorem implies that there exists a real number c in (a, b) with $f'(c) = 0$. But this contradicts our hypothesis about the derivative not being zero anywhere in (a, b). We conclude that there can be at most one point p in (a, b) with $f(p) = 0$.

- **11**: $P_2(x) = \frac{f(0)}{0!}x^0 + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2$. We have $f'(x) = e^x \cos x - e^x \sin x$, and $f''(x) = e^x \cos x - e^x \sin x - (e^x \sin x + e^x \cos x) = -2e^x \sin x$. So $f(0) = 1$; $f'(0) = 1$; and $f''(0) = 0$. Thus

$$P_2(x) = 1 + x$$

  a. Use $P_2(0.5)$ to approximate $f(0.5)$. Find an upper bound for error $|f(0.5) - P_2(0.5)|$ using the error formula, and compare it to the actual error.

  Solution: $P_2(0.5) = 1.5$. The error formula tells us that $|f(0.5) - P_2(0.5)|$ is bounded by

$$\left| \frac{f'''(\xi)}{3!}(0.5)^3 \right|$$

  where ξ is some number between 0 and 0.5. We have $f'''(x) = -2e^x \sin x - 2e^x \cos x = -2e^x(\sin x + \cos x)$; since $\sin x + \cos x$ is bounded in absolute value by 2, and $-2e^x$ is bounded in absolute value by $2\sqrt{e}$ for $x \in [0, 0.5]$, an upper bound on the error is $4\sqrt{e}/48 \approx 0.1374$. (With a bit more work, one can show that $|\sin x + \cos x| \leq \sqrt{2}$, which gives the stronger bound $2\sqrt{2e}/48 \approx 0.09715$. Yet another approach is to show that $e^x(\sin x + \cos x)$ is increasing on $[0, 0.5]$, which gives the still stronger bound

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$2\sqrt{e}(\sin(0.5)+\cos(0.5))/48 \approx 0.09322$. Any of these answers is acceptable.) The actual error is $|1.446889 - 1.5| = .05311$

b. Find a bound for the error $|f(x)-P_2(x)|$ using $P_2(x)$ to approximate $f(x)$ on the interval $[0,1]$.

Solution: Here we need to determine an upper bound for $|\frac{-2e^{\xi}(\sin\xi+\cos\xi)}{6}\cdot x^3|$ for $0 < \xi < x \le 1$. Again $|\sin x + \cos x|$ is bounded by 2 (or by $\sqrt{2}$) and $|e^x|$ is bounded by e for $x \in [0,1]$. From these estimates we get an upper bound of $|f(x) - P_2(x)| \le \frac{4e}{6}x^3$ (or $\frac{2e\sqrt{2}}{6}x^3$) for $x \in [0,1]$. The answer in the book comes from $e^{\xi}(\sin\xi + \cos\xi)$ being increasing on $[0,1]$, and using 1 as an upper bound on x^3: $2e(\sin(1)+\cos(1))/6 \approx 1.252$.

c. Approximate $\int_0^1 f(x)dx$ using $\int_0^1 P_2(x)dx$.

Solution: $\int_0^1(1+x)dx = \left[x + \frac{x^2}{2}\right]_0^1 = 1.5$.

d. Find an upper bound for the error in (c) using $\int_0^1 |R_2(x)|dx$ and compare the bound to the actual error.

Solution: From (b), we have $|R(x)| \le \frac{4e}{6}x^3$. Integrating, this yields $\frac{e}{6}x^4\big|_0^1 \approx 0.453$. (Other acceptable solutions are $\int_0^1 \frac{2e\sqrt{2}}{6}x^3dx \approx 0.320$ and $\int_0^1 1.252\,x^3\,dx \approx 0.313$.)

To find the actual error, we compute $\int_0^1 e^x\cos x\,dx = \left[\frac{1}{2}e^x(\sin x + \cos x)\right]_0^1 = \frac{e}{2}(\cos 1 + \sin 1) - \frac{1}{2} \approx 1.37802$. So the actual error is the difference between this and 1.5, which is about 0.1220.

- **21**: Using the Taylor expansion for $\cos x$ about $x_0 = 0$, (and following the example in the textbook), we have $\cos x = 1 - \frac{1}{2}x^2 + \frac{1}{24}x^4\cos\xi(x)$ where $\xi(x)$ is between 0 and x. The maximum error is the maximum possible value of $|\frac{1}{24}x^4\cos\xi(x)|$ for $x \in \left[-\frac{1}{2},\frac{1}{2}\right]$. We know $\cos\xi(x)$ is always in $[-1,1]$, and the most $|x^4|$ can be for x in this interval is $1/2^4 = \frac{1}{16}$. So a bound on the maximum error is $\frac{1}{24\cdot 16} = 0.002604$.

- **24**: The *error function* defined by $\left(\text{erf}(x) = \frac{2}{\sqrt{\pi}}\int_0^x e^{-t^2}\,dt\right)$ gives the probability that any one of a series of trials will lie within x units of the mean, assuming that the trials have a normal distribution with mean 0 and standard deviation $\sqrt{2}/2$. This integral cannot be evaluated in terms of elementary functions, so an approximating technique must be used.

  a. Intergrate the Maclaurin series for e^{-x^2} to show that $\text{erf}(x) = \frac{2}{\sqrt{\pi}}\sum_{k=0}^{\infty}\frac{(-1)^k x^{2k+1}}{(2k+1)k!}$

  Solution: We know $e^y = \sum_{k=0}^{\infty} y^k/k!$. Thus $e^{-x^2} = \sum_{k=0}^{\infty}(-1)^k x^{2k}/k!$. Integrating term by term gives the desired result.

  c. Use the series in part (a) to approximate erf(1) to within 10^{-7}.

  Solution: When $x = 1$, the series in part (a) has the form $\sum_{k=0}^{\infty}(-1)^k a_k$ with a_k positive and $a_{k+1} < a_k$. (Note that the ratio $a_{k+1}/a_k = \frac{2k+1}{2k+3}\cdot\frac{k}{k+1}\cdot x^2$, which is less than 1.) The alternating series test guarantees that the error in truncating the

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series is bounded by the first omitted term. Using matlab (or any computer algebra system such as Wolfram alpha), we find that $\frac{2}{\sqrt{\pi}} \cdot \frac{1}{(2k+1)k!} = 1.64 \times 10^{-7}$ when $k = 9$ and 1.48×10^{-8} when $k = 10$. Thus, we can omit the terms $k \geq 10$. Our approximation is then $\operatorname{erf}(1) \approx \frac{2}{\sqrt{\pi}} \sum_{k=0}^{9} \frac{(-1)^k}{(2k+1)k!} = 0.842700779$. (Including more terms yields 0.842700793).

Alternative solution: We can use Taylor's theorem with remainder to bound the error. Let $\tilde{f}(y) = e^y = \tilde{P}_n(y) + \tilde{R}_n(y)$, where $\tilde{P}_n(y) = \sum_{k=0}^{n} \frac{1}{k!} y^k$ and $\tilde{R}_n(y) = \frac{1}{(n+1)!} e^{\xi(y)} y^{n+1}$. Then $f(x) = e^{-x^2}$ is equal to $\tilde{f}(-x^2)$. It follows that $P_{2n+1}(x) = \tilde{P}_n(-x^2)$ and $R_{2n+1}(x) = \tilde{R}_n(-x^2)$. Note that $|R_{2n+1}(x)| = \left|\frac{e^{\xi(-x^2)}}{(n+1)!}(-1)^{n+1} x^{2n+2}\right| \leq \frac{1}{(n+1)!} x^{2n+2}$, since e^x is an increasing function and $\xi(-x^2)$ is between $-x^2$ and 0 (so $e^{\xi(-x^2)} \leq e^0 = 1$.) Integrating from 0 to 1, we have

$$\left| \operatorname{erf}(1) - \frac{2}{\sqrt{\pi}} \sum_{k=0}^{n} \frac{(-1)^k 1^{2k+1}}{(2k+1)k!} \right| \leq \frac{2}{\sqrt{\pi}} \int_0^1 |R_{2n+1}(x)|\, dx \leq \frac{2}{\sqrt{\pi}} \frac{x^{2n+3}}{(2n+3)(n+1)!} \Bigg|_0^1.$$

The bound is identical to the first approach (i.e. $n = 9$ suffices).

- **28:**

  a. Show that if f satisfies a Lipschitz condition with Lipschitz constant L on $[a, b]$, then $f \in C[a, b]$.

  Solution: Increasing L if necessary, we may assume $L > 0$. Let $x_0 \in [a, b]$ and $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{L}$. Suppose $x \in [a, b]$ with $|x - x_0| < \delta$. Then $|f(x) - f(x_0)| \leq L|x - x_0| < L \cdot \delta = \epsilon$. So f is continuous at x_0, by definition. Since x_0 was arbitrary, $f \in C[a, b]$.

  b. Show that if f has a derivative that is bounded on $[a, b]$ by L, then f satisfies a Lipschitz condition with Lipschitz constant L on $[a, b]$.

  Solution: Assume that $|f'(x)| \leq L$ for every $x \in [a, b]$. Let $x, y \in [a, b]$. Our assumptions imply that f is differentiable on (a, b) so by the mean value theorem there exists a constant $c \in [x, y]$ such that $f'(c) = \frac{f(x) - f(y)}{x - y}$. Rewriting this as $|f(x) - f(y)| = f'(c) \cdot |x - y|$ and applying the bounded assumption, we obtain $|f(x) - f(y)| \leq L|x - y|$. So f satisfies a Lipschitz condition with Lipschitz constant L on $[a, b]$.

  c. Give an example of a function that is continuous on a closed interval but does not satisfy a Lipschitz condition on the interval.

  Solution: One example is $f(x) = \sqrt{x}$ on $[0, 1]$. This function is continuous on $[0, 1]$ but if you consider x, y pairs of the form $0, \epsilon$ for small ϵ, then the ratio $\frac{|f(x) - f(y)|}{|x - y|} = \left|\frac{\sqrt{\epsilon}}{\epsilon}\right| = \left|\frac{1}{\sqrt{\epsilon}}\right|$ becomes increasingly large, and cannot be uniformly bounded by any constant L. Thus f does not satisfy any Lipschitz condition on this interval. [Notice that the derivative of f is not bounded when x nears zero — it follows from part (b) that this is a necessary condition for a differentiable example.]

- **29:**

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b. Suppose that c_1 and c_2 are positive constants. Show that a number ξ exists between x_1 and x_2 with $f(\xi) = \frac{c_1 f(x_1)+c_2 f(x_2)}{c_1+c_2}$.

Solution: Relabeling x_1 and x_2 if necessary, we may assume $f(x_1) \leq f(x_2)$. Then

$$f(x_1) = \frac{c_1+c_2}{c_1+c_2} f(x_1) \leq \frac{c_1 f(x_1) + c_2 f(x_2)}{c_1+c_2} \leq \frac{c_1+c_2}{c_1+c_2} f(x_2) = f(x_2),$$

where we used $c_2 f(x_1) \leq c_2 f(x_2)$ and $c_1 f(x_1) \leq c_1 f(x_2)$, which requires $c_1 \geq 0$, $c_2 \geq 0$. By the intermediate value theorem, there is a ξ between x_1 and x_2 such that $f(\xi) = \frac{c_1 f(x_1)+c_2 f(x_2)}{c_1+c_2}$, as claimed.

c. Give an example to show that the result in part (b) does not necessarily hold when c_1 and c_2 have opposite signs with $c_1 \neq c_2$.

Solution: Let $c_1 = -1$, $c_2 = 2$, $f(x) = x$, $a = -1$, $b = 2$, $x_1 = 0$, $x_2 = 1$. Then for $\xi \in [x_1, x_2]$, $0 \leq f(\xi) \leq 1$, which is always smaller than $\frac{c_1 f(x_1)+c_2 f(x_2)}{c_1+c_2} = 2$.

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